

EE/CprE/SE 4920 STATUS REPORT 2

2/5/26 - 2/12/26

Group Number: sdmay26-40

Project Title: True Force Technologies Arcade Rack

Client: True Force Technologies

Advisor: Matt Post

Team Members/Role:

Jacob Garcia - Frontend Development

Dylan Longlett - Frontend Development

Andy Drafahl - Team Manager, Frontend Development

Fadi Masannat - Frontend Development

Sofi Gutierrez - Backend Development

Parnika Dasgupta - Backend Development

Colin Yuska - Electronics

Weekly Summary

Thanks to the enthusiasm of one of our group members (shout out Dylan), the group is way ahead of schedule. Not only is the screen on the arcade cabinet fixed and fully functional, but the joystick and play button have been successfully wired up as well. 3 minigames on the arcade cabinet are pretty much finished, with several more on the way. The arcade cabinet is ready for user testing in the TLA and will soon be ready for testing at different locations. The team also added documentation and performed additional research on the cabinet itself.

Past Week Accomplishments

Jacob Garcia

Began work on developing a new arcade mini game for the arcade cabinet.

Parnika Dasgupta

I have started working on the code for Flappy Bird with Sofi. I have been going through her code and understanding it. I have been putting comments around her code for everyone to be able to understand what is going on. I am currently working on trying to make the game more fun.

Dylan Longlett

New Arduino, button, joystick, LED, and Pi setup content, with code and instructions under ArduinoCode, ButtonControl, LEDControl, and PiRelatedInfo.

A complete game selector experience has been added under GameSelector with fonts and game images for the arcade UI.

Two full game implementations were added: KingMaker and PowerPuller3000 u, including gameplay logic, leaderboard management, wiring/Arduino integration, and audio assets.

Shared wiring helpers and install guides added for power/weight hardware under WeightConnection and WIRED_INSTALL.md.

Name-entry JSON configuration added at name_entry.json.

Large asset drops across the arcade stack: fonts, images, and audio files added under the various game folders and selector folders, plus an audio editor installer under audios.

Fadi Masannat

I have researched our codebase and existing documentation to create six comprehensive technical and detailed visuals and diagrams relevant to our data and user flows and architecture. Additionally, I have updated our documentation to a great comprehensive place by integrating the Python Arcade mode fully into the documentation. Structured it for the rest of the team for good exposure.

Sofi Gutierrez

I started to clean the backend and reorganize to to the new direction of this project. There were repositories and controllers that we will not use that the old backend had. I also started to create repositories and controllers for the minigame,

Colin Yuska

Finished collecting resources on arcade maintenance and modification. In addition found projects that will be useful as a guide for potential new features.

Andy Drafahl

I chose a new service for the team to use for keeping track of task assignments and set it all up for the group, only to realize that at this point, we're probably better off without it. I didn't make as much progress this week as I would have liked, and plan to pick up the pace in the coming week.

Individual Contributions

Team Member 1: Worked on... Team Member 2: Team Member 3: ... o Individual contributions (Creating this section is optional, but it is Required to include the "Hours Worked for the Week" and their "Total Cumulative Hours" for the project for each member somewhere relevant in your report. Your individual weekly hours should be at a minimum of 6-8 hours for this course. So please manage your time well. Also, ensure that individual contributions support your claim to the weekly hours. Be honest with the reports. **DELETE THIS DESCRIPTION WHEN FINISHED]**

Name	Individual Contributions	Hours This Week	Hours Cumulative
Jacob Garcia	Begin development on a new arcade game for machine.	3	5
Parnika Dasgupta	Working on Flappybird	1	3
Dylan Longlett	New Arduino, button, joystick, LED, and Pi setup content, with code and instructions under ArduinoCode, ButtonControl, LEDControl, and PiRelatedInfo. A complete game selector experience added under GameSelector with fonts and game images for the arcade UI. Two full game implementations added: KingMaker and PowerPuller3000 And lots of wiring	25 (Note from team manager: This number is no exaggeration)	32.67
Sofi Gutierrez	Worked on the backend to start matching up with the new direction of the project	1	4
Colin Yuska	Finished research for resources to modify and cleanup our cabinets design.	2	2.75
Fadi Masannat	Created capstone ready comprehensive technical and detailed visuals and diagrams. Updated the documentation to	3	7

	a great comprehensive place by integrating the python Arcade mode fully into the documentation. Structured it for the rest of the team for good exposure.		
Andy Drafahl	Chose a new service for team tasks, set up all of the directories, and pushed it to the team, though we decided to move away from a kanban board anyway.	.3	2.33

Plans for the Upcoming Week

Jacob Garcia

I will be continuing development on my arcade game and plan on having a full prototype by next week.

Fadi Masannat

I will be experimenting with the Python arcade app and possibly build another unique minigame. Or try to find bugs and harden the current codebase.

Sofi Gutierrez

Depending on where I am needed, the main focus may change, but I will continue working on the minigame. I would like to get the backend working sooner, so depending on the work for the minigame needed, I may shift my focus to the backend

Colin Yuska

Clean up the mess of wires within the cabinet, and complete adding the receipt printer to the exterior.

Andy Drafahl

Make final adjustments to the animation doc and send it to the animator (finally). Work on music and recover lost hours.

Parnika Dasgupta

I will keep working on Flappy Bird and make it more fun and attractive to the players.

Summary of Weekly Advisor Meeting

Our advisor is out with an injury, but we showed our client the arcade cabinet with several minigames fully functioning. In his words, he was "blown away." There's not much else to say; we're in a good spot.